Woodham
Science

Year 8
Science Revision Guide

Name: _____

How to use this booklet

This booklet contains lots of information to help you to revise for your end of year assessment. You will find the key points you need to learn and some practice questions/tasks for each unit. You should attempt the practice questions and then check your answers with the key information provided. You should keep attempting the practice questions at regular intervals until you can regularly remember all answers.

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Biology Unit 1: Cells

Key points to learn:

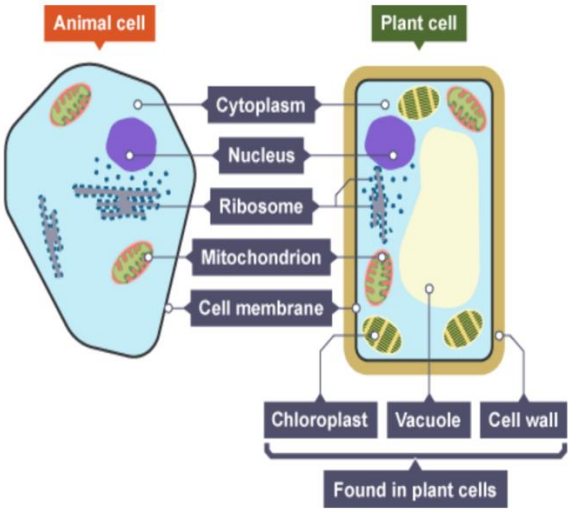
1. How to label an animal cell.
2. How to label a plant cell.
3. How to label a prokaryotic (bacterial) cell.
4. The function of each sub-cellular structure.
5. The different parts of a microscope.
6. How to use a microscope.
7. How to calculate magnification.
8. The differences between plant and animal cells.
9. The differences between prokaryotic and eukaryotic cells.

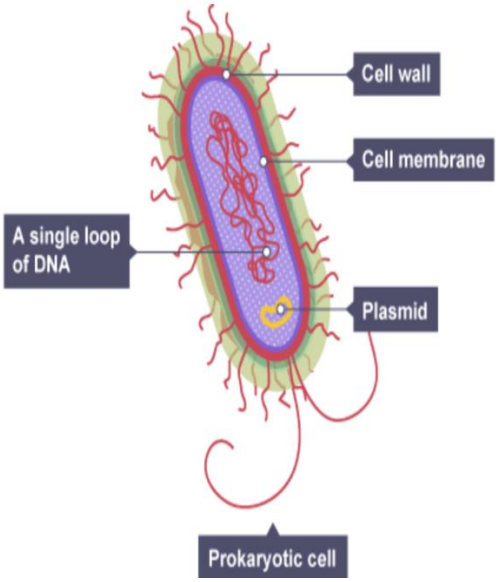
Practice Tasks/Questions

1. Draw and label an animal cell.
2. Draw and label a plant cell.
3. Write down the functions of the nucleus, cell membrane, cytoplasm, ribosomes, mitochondria, vacuole, cell wall and chloroplast.
4. Write down the 3 sub-cellular structures found in plant cells but not found in animal cells.
5. Prokaryotic cells have a nucleus – True or false?
6. Complete the equation for magnification:
   Image size = Actual size x _____
7. When would you change the objective lens to a higher magnification when using a microscope?
8. When would you rotate the focussing wheel when using a microscope?

Biology Unit 1: Cells

Key information to learn:





Sub-cellular structure	Function
Nucleus	Controls the activities of the cell
Cell membrane	Controls what goes in and out of the cell
Cytoplasm	Where most chemical reactions take place
Ribosomes	Where proteins are made
Mitochondria	Where aerobic respiration takes place
Vacuole	Stores cell sap
Cell Wall	Give cells structure and support
Chloroplast	Where photosynthesis takes place



Microscope key points

- If the image is too small, change the objective lens to a higher magnification.
- If an image is blurry, rotate the focussing knob until the image is in focus.
- If an image is too dark, adjust the mirror to add light to the sample.
- Image size = Actual Size x Magnification

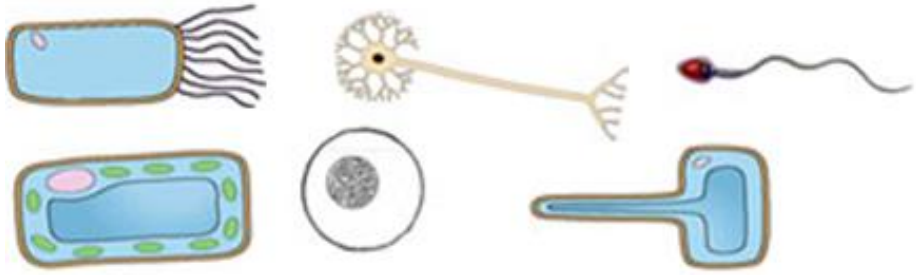
Biology Unit 2: Organisation

Key points to learn

1. How to label a specialised animal cell.
2. How to label a specialised plant cell.
3. Examples of specialised cells in animals and plants.
4. The four components of blood.
5. The names of the three types of blood vessel.
6. The function of the three types of blood vessel.
7. The main parts of the heart.
8. The parts of the digestive system.
9. The pathway the food takes through the digestive system
10. The role of enzymes in the digestive system.

Practice Tasks/Questions

1. Draw and label a specialised animal cell.
2. Draw and label a specialised plant cell.
3. Write down examples of specialised cells in animals and in plants.
4. a) What 3 structures do all these cells have?
   b) Which two cells are plant cells? What other structures do they have?
   c) Name each cell

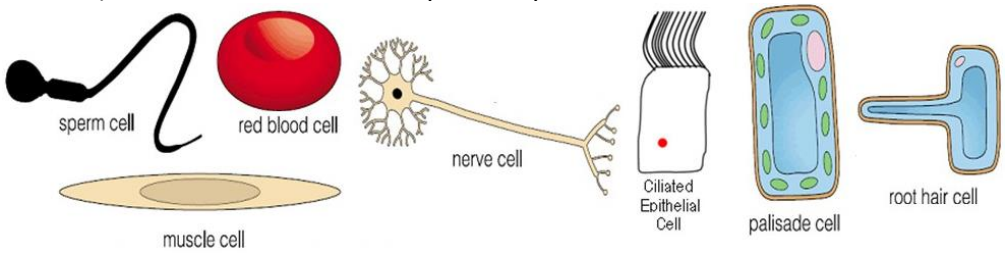


5. State the four main components of blood and what they do.
6. What are the three main types of blood vessel?
7. What does each blood vessel do?
8. a) What are the names of the top two chambers of the heart?
   b) What are the names of the bottom two chambers of the heart.
   c) What type of blood vessel takes blood into the heart?
   d) What type of blood vessel takes blood out of or away from the heart?
9. What are the main organs in the digestive system?
10. Outline the pathway the food takes through the digestive system.
11. How do enzymes help in digestion?

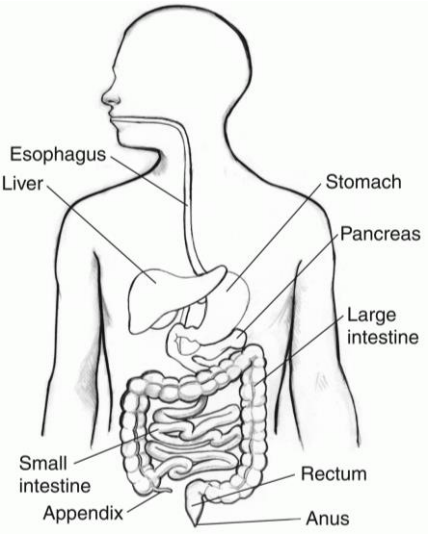
Biology Unit 2: Organisation

Key information to learn:

Specialised cells - Root hair cells and palisade cells are examples of plants cells (they both have a cell wall). All the others are examples of specialised animal cells.

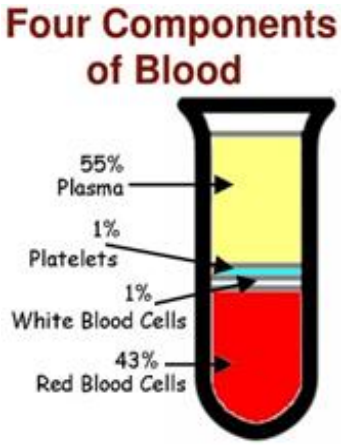


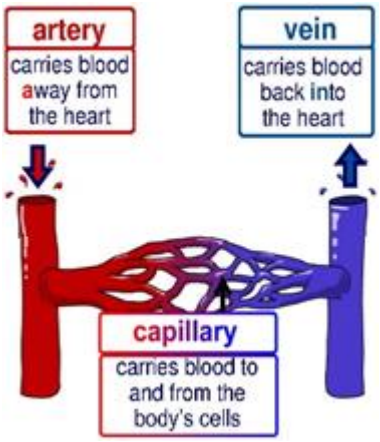
The digestive system: breaks down food and absorbs it into the blood

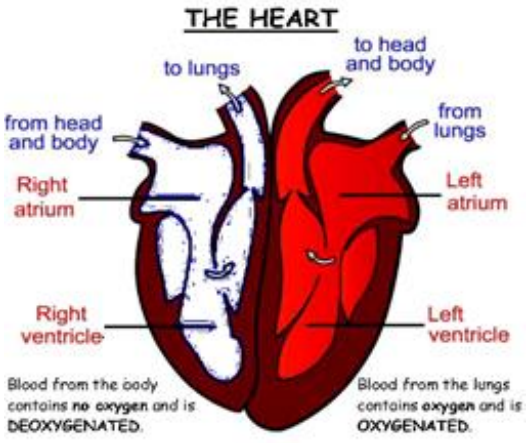


- The food is chewed in the **mouth** and mixed with saliva
- The chewed food then moves down the **oesophagus** to the stomach
- In the **stomach**, the food is mixed with acid (which kills microbes) and other chemicals
- The mixture then passes into the **small intestine** where it is mixed with more chemicals (enzymes and bile).
- *The nutrients are absorbed into the blood*
- In the **large intestines** water is removed and the waste is then stored in the **rectum** until it can be removed, through the **anus**
- Enzymes are very special chemicals that help to break the food down more quickly.

The circulatory system: blood, blood vessels and the heart – transports blood around the body.







Plasma	Liquid part of the blood. Transport nutrients
Platelets	Help to clot the blood
White blood cells	Help to fight organisms causing disease (pathogens)
Red blood cells	Transport oxygen around the body to the cells

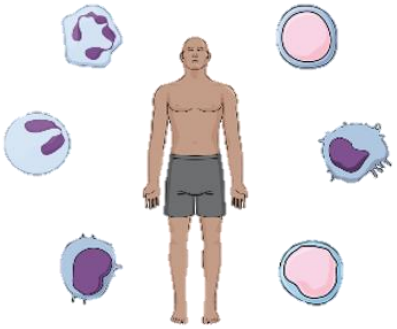
Biology Unit 3: Infection and Response

Key points to learn

1. Describe health.
2. Factors that can affect health.
3. The description of a 'pathogen'.
4. The four microorganisms that can be a pathogen.
5. The ways that disease/pathogens can be spread.
6. How the body can prevent microbes getting into the blood.
7. How the white blood cells can destroy microbes if they do get in to the blood.

Practice Tasks/Questions

1. State the definition of the term 'health'.
2. State two factors that can affect heath.
3. What is a 'pathogen'?
4. State the four microorganisms that can be pathogens.
5. Name three ways that pathogens can spread disease.
6. State three ways that the body prevents pathogens from entering the body?
7. Describe how the white blood cells can destroy pathogens.
8. The main function of white blood cells (WBCs) in the body is to:



A – carry oxygen
B – Help in clot formation
C – Produce more red cells
D – protect body against pathogens

Biology Unit 3: Infection and Response

Key information to learn:

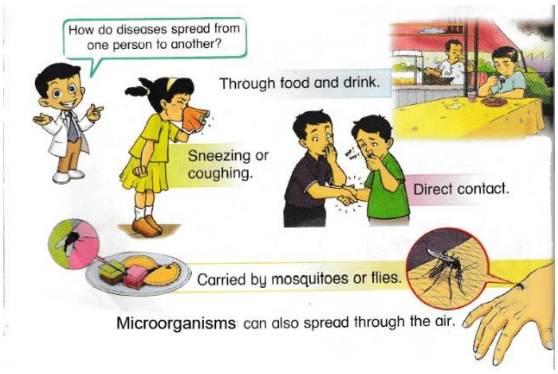
Health:
Health is "a state of complete physical, mental, and social well-being and not merely the absence of disease"

Physical is about the body. Mental is about how people think and feel.
Social well-being is the extent to which you feel a sense of belonging.

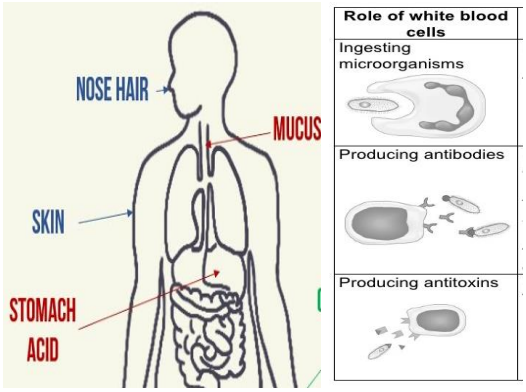
Pathogens: a microorganism that can cause disease (viruses are the **smallest**)

Type of pathogen	Description	Human diseases caused by pathogens of that type
Bacteria *Escherichia coli* 	Single-celled organisms without a nucleus	Strep throat, staph infections, tuberculosis, food poisoning, tetanus, pneumonia, syphilis
Viruses *Herpes simplex* 	Thread-like particles that reproduce by taking over living cells	Common cold, flu, genital herpes, cold sores, measles, AIDS, genital warts, chiken pox, small pox
Fungi *Death cap mushroom* 	Simple organisms, including mushrooms and yeasts, that grow as single cells or thread like filaments	Ringworm, athlete's foot, tinea, candidiasis, histoplasmosis, mushroom poisoning

Spreading pathogens:



Body responses to pathogens:



Facts:
Pathogens can be spread by air (sneezing), by contact (touching) and by water (if it is contaminated)
The body will try and stop pathogens getting into the blood – the skin acts as a barrier, the nose has hairs and mucus that trap the pathogens and the stomach has acid which will kill the pathogens.
If the pathogens do get into the blood then the **white blood cells** will try and destroy them by **ingesting** them or producing **antibodies** and **antitoxins**.

Biology Unit 4: Bioenergetics

Key points to learn

1. What do we mean by photosynthesis?
2. What are the reactants and products of photosynthesis?
3. How do plants get the reactants of photosynthesis?
4. How can we test whether photosynthesis has happened?
5. What do we mean by respiration?
6. How is respiration different from breathing?
7. What is aerobic respiration?
8. What is anaerobic respiration?
9. What are the differences between anaerobic respiration in animals and in plants/microorganisms?

Practice Tasks/Questions

1. Explain what photosynthesis is and write down the word equation for the reaction.
2. How do plants transport the water from the soil to the leaves for photosynthesis?
3. How does the starch test show whether photosynthesis has occurred and what does a positive result look like?
4. Explain what the difference between respiration and breathing (ventilation) is.
5. Explain what aerobic respiration is and write down the equation for the reaction.
6. Explain what anaerobic respiration is and how it differs from aerobic respiration.
7. Compare and contrast anaerobic respiration in plants/microorganisms and animals, including noting down the equations.

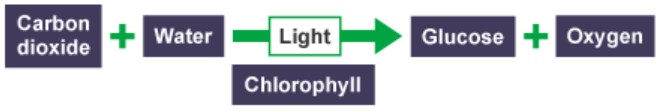
Biology Unit 4: Bioenergetics

Key information to learn:

Keyword	Definition
Reactant	The substances go into the reaction, these are a bit like the ingredients.
Product	The substances made in the reaction.
Photosynthesis	A chemical reaction in plants where they make glucose and oxygen from carbon dioxide and water using light energy.
Chlorophyll	A green pigment/chemical found in chloroplasts of plant cells.
Stomata	Small holes on the bottom of a leaf which allow carbon dioxide to enter and oxygen to leave.
Roots	Organs of plant that are underground which keep the plant anchored and absorb water and nutrients.
Xylem	Tube found in the stem which transport water and minerals to the leaf.
Respiration	A chemical reaction in which energy is released from glucose (sugar).
Ventilation	The process of breathing air into and out of the lungs.
Aerobic Respiration	The type of respiration which needs oxygen.
Anaerobic Respiration	The type of respiration which does not need oxygen.
Fermentation	Anaerobic respiration in yeast cells. It is used to make beer, bread and wine.

Photosynthesis

Photosynthesis is the process in which plants make their



The leaves of the plant are the organ at which photosynthesis happens.

The reactants of photosynthesis are water and carbon dioxide. The carbon dioxide enters the leaf through the stomata. The water enters the plant through the roots and is transported up the xylem to the leaf.

Light energy allows the reaction to occur but is not a reactant. It is absorbed by a green pigment called chlorophyll which is found in the chloroplasts in the cells of the leaves.

A variegated leaf is one which has white parts and green parts. The white parts have no chloroplasts and so cannot photosynthesise.

The plant may convert some of the glucose made in photosynthesis to starch so it can be stored for later use. We can test for whether photosynthesis has happened by testing for starch using iodine. If the leaf turns blue/black, then starch has been made which means photosynthesis has happened.

Respiration

Respiration is a chemical reaction which releases energy from glucose (sugar). It is not ventilation which is the process of breathing in and out.

The energy released in respiration is needed for many things, e.g. movement and other chemical reactions needed to keep living organisms alive.

Aerobic Respiration

This is the type of respiration which releases energy from glucose by combining it with oxygen. It happens in the mitochondria within cells.

glucose + oxygen → carbon dioxide + water + energy

Anaerobic Respiration

This is the type of respiration which does not require oxygen but releases less energy.

Anaerobic Respiration in Animals

glucose → lactic acid + energy

Anaerobic Respiration in Yeast (Fermentation)

glucose → ethanol + carbon dioxide + energy

Biology Unit 5: Reproduction and Variation

Key points to learn

1. To explain what reproduction is and to identify the cells involved in human reproduction.
2. To label and explain the functions of the human reproductive system.
3. To explain what fertilisation is and the processes involved.
4. To explain what happens during gestation and birth.
5. To explain what puberty is and identify the changes which occur to males, females and both.
6. To explain what happens during the menstrual cycle.
7. To explain what contraception is used for and list several methods of contraception.
8. To explain how plants can reproduce and label the reproductive system of the plant.
9. To explain what is meant by variation and how variation can be caused.

Practice Tasks/Questions

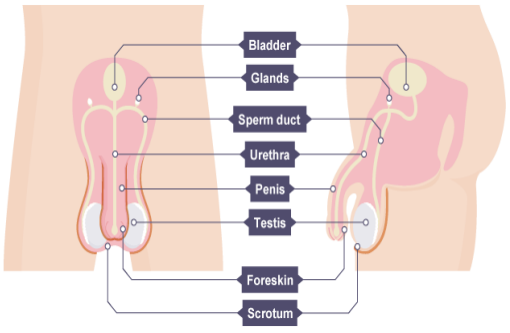
1. What is the male reproductive cell and how is it adapted to its job?
2. What is the function of the ovaries and the uterus in the female reproductive system?
3. What is meant by fertilisation and where does it happen?
4. What is the difference between an embryo and fetus?
5. What is the function of the amniotic fluid in gestation?
6. What happens during the first stage of the menstrual cycle?
7. Explain how condoms prevent both pregnancy and STIs.
8. What is the function of the anther in plant reproduction?
9. What do we mean by the term ‘variation’ and what are its possible causes?
10. List two characteristics that show environmental variation.

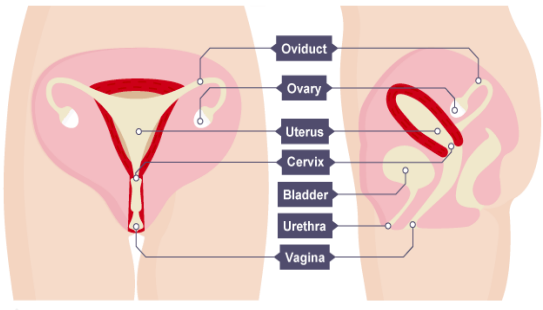
Biology Unit 5: Reproduction and Variation

Key information to learn:

Keyword	Definition
Reproduction	The creation of new members in a species to prevent extinction and pass down genes.
Sperm Cell	The male sex cell which is produced by the testes in animals.
Egg Cell (Ovum)	The female sex cell which is produced by the ovaries in animals.
Fertilisation	The fusing of the sperm and the egg cell. This happens in the oviduct.
Implantation	The sinking of a fertilised egg into the lining of the uterus (womb).
Gestation	The time during which a fertilised egg develops into a baby ready to be born. This happens in the uterus (womb).
Embryo	An unborn baby before the 8th week of gestation in humans.
Fetus	An unborn baby after the 8th week of gestation in humans.
Puberty	Time during which sexual maturity happens.
Menstruation	The loss of blood and tissue from the uterus through the vagina during the first stage of the menstrual cycle. It is also called a period.
STI	A sexually transmitted infection, e.g. genital herpes, chlamydia, and gonorrhoea.
Pollination	The fertilisation of flowers by passing on their pollen.
Variation	The differences in characteristics between members of the same species or between members of different species.

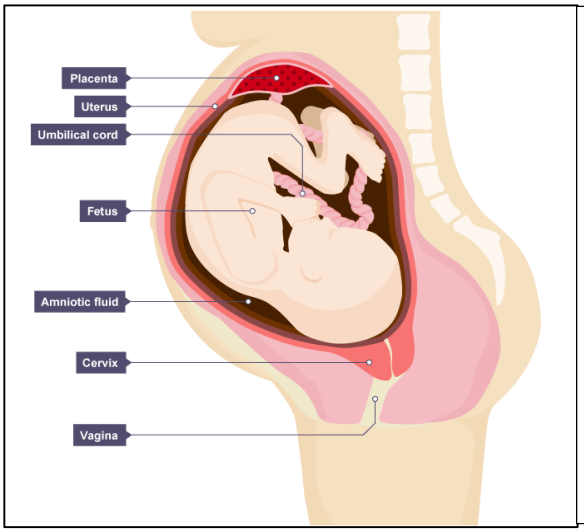
Structure	Male or Female	Function
Testis	Male	Produces sperm cells
Scrotum	Male	Skin which protects the testis
Sperm Duct	Male	Tube which the sperm travels out of the testis in.
Glands	Male	Produces nutrient-rich fluid that mixes with the sperm cells to produce semen.
Penis	Male	Carries semen and urine out the body.
Ovary	Female	Produces egg cells (ova).
Oviduct	Female	Tubes which connect ovaries to uterus (womb).
Uterus	Female	This is where the fertilised egg develops into a baby that is ready to be born.
Cervix	Female	Ring of muscle at bottom of uterus which keeps baby in place during pregnancy.
Vagina	Female	Muscular tube which leads from the cervix to outside of the body.





Fertilisation is where the sperm and the egg fuse together, it happens in the oviducts. The fertilised egg will then travel down the oviduct to the uterus where implantation happens.

If implantation does not occur, then women will experience menstruation. If implantation does occur, then gestation begins which lasts for 40 weeks/9 months in humans.



Gestation and Birth

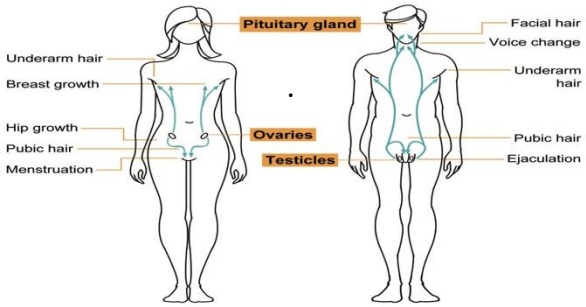
The unborn baby is protected by the uterus and the amniotic fluid.

The placenta is the organ responsible for providing oxygen and nutrients and removing waste substances. It grows into the wall of the uterus and is connected to the baby by the umbilical cord.

During birth, the cervix relaxes and muscles in the wall of the uterus contract. Waves of muscle contraction push the baby out of the mother's body through the vagina

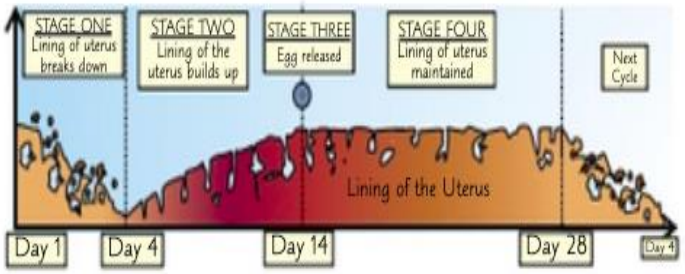
Puberty

Puberty happens to boys and girls between the ages of 8-14



The Menstrual Cycle

The menstrual cycle is a monthly sequence of events which females go through from puberty onwards. It involves the body preparing the uterus in case it receives a fertilised egg.



Contraception

Contraception is something that is used to prevent pregnancy from occurring during sexual intercourse. Some forms of contraception may also prevent sexually transmitted infections (STIs) if they create a barrier (e.g. condoms).

There are mechanical, chemical, surgical and natural contraceptive methods.

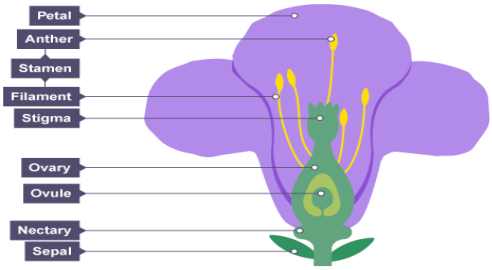
Variation

The differences in characteristics between members of the same species or between members of different species.

These can be caused by genes (e.g. eye colour and blood group), environment (e.g. piercings and accent) or a mixture of both genes and environment (e.g. height and mass).

Plant Reproduction

Structure	Function
Sepals	Protect the unopened flower
Petals	May be brightly coloured to attract insects
Stamens	The male parts of the flower (each consists of an anther held up on a filament)
Anthers	Produce male sex cells (pollen grains)
Stigma	The top of the female part of the flower which collects pollen grains
Ovary	Produces the female sex cells (contained in the ovules)
Nectary	Produce a sugary solution called nectar, which attracts insects



Biology Unit 6: Ecology

Key points to learn:

1. The definitions of the following key terms: Habitat, adaptation, community, ecosystem, population, environment, producer, consumer, predator, prey, omnivore, herbivore, carnivore, photosynthesis, interdependence.
2. How to draw a food chain.
3. How a polar bear is adapted to being a hunter.
4. How a camel is adapted to live in the desert.
5. What happens during decay.
6. Why decay is important.
7. How humans can harm the environment.
8. Ways humans can try to reduce their effect on the environment.

Practice Tasks/Questions

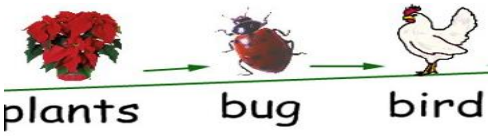
1. What is the difference between an ecosystem and a community?
2. Draw a simple food chain including a producer, primary consumer and a secondary consumer.
3. Explain the difference between an omnivore, herbivore and a carnivore.
4. How does having a hump for storage help a camel to survive in the desert?
5. A polar bear has a white colour that helps it blend in with the snow. What is it called when an organism's colour blends in with the surroundings?
6. Plants get nutrients from the soil. How do the nutrients get to the soil originally?
7. State 3 ways that humans are harming the environment.

Biology Unit 6: Ecology

Key information to learn:

Keyword	Definition
Habitat	The place an organism lives
Adaptation	The features plants and animals have to help them to survive
Community	All the plants and animals that live in a habitat
Ecosystem	A community of animals and plants, together with the habitat where they live
Population	All the members of a single species living in a habitat
Environment	All the conditions that surround any living organism
Producer	Green plants that produce their own food
Prey	Animals that consume (eat) other living organisms (plants or animals).
Omnivore	An organism that eats plants and animals
Herbivore	An organism that easts only plants
Carnivore	An organism that eats only animals
Photosynthesis	The process by which green plants make their own food
Interdependence	Organisms depending (needing) other organisms to survive.

- The arrows in a food chain show the way in which energy is moving. The food chain below shows the energy of the plant being transferred to the bug. From the bug, the energy is then transferred to the bird.



- The table below summaries the adaptations of camels and polar bears.

Camel	Polar Bear
1, Hump for storing food 2, Long eyelashes to keep sand out of eyes 3, Padded feed to prevent sinking into the ground.	1, Strong muscles for over-powering and catching prey 2, White colour to camouflage 3, Sharp claws and teeth.

- Plants absorb the nutrients from the soil. The nutrients get into the soil due to decay. Dead organisms and waste are decayed, releasing the nutrient into the soil for the plants to use.
- Without decay, there would be no recycling of nutrients in the environment.
- Humans are having a negative impact on the environment due to the rapid increase in population and the increased standard of living. For example, humans have had the following effects: Increase in air, water and land pollution, deforestation, and global warming.
- Humans have tried to lessen the impact with the following programmes: Breeding programmes for endangered species, protection of rare habitats, reintroduction of field margins and hedgerows, reduction of deforestation and carbon emissions.

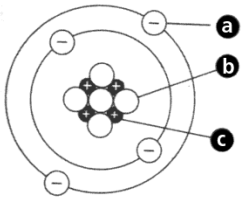
Chemistry Unit 1: Atomic Structure

Key points to learn:

1. What an element is.
2. What a compound is.
3. What a mixture is.
4. What an atom is.
5. What parts an atom contains.
6. What the symbols of the periodic table represent.
7. What the different parts of the periodic table are.
8. How do groups such as group one react.

Practice Tasks/Questions

1. Define an element.
2. Define a compound.
3. Define a mixture.
4. Label the image of an atom.

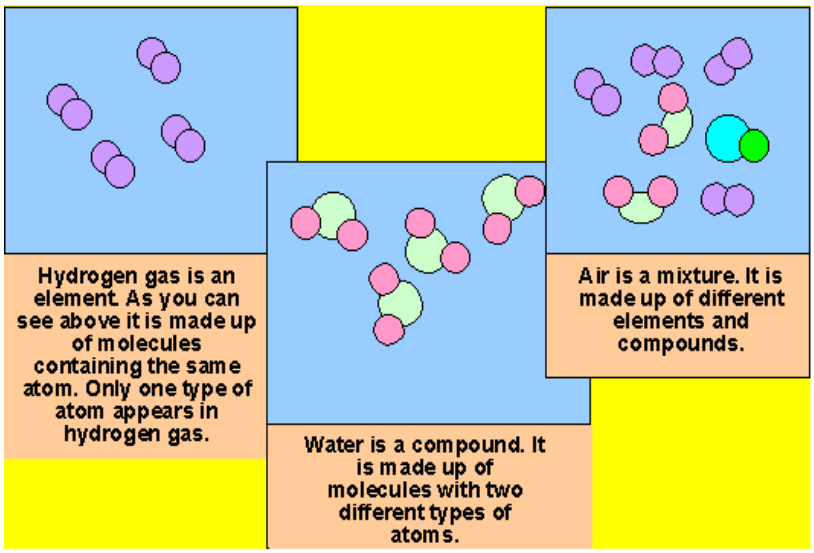


5. Find the symbol for the following element.
   O = N = C = H = Cl =
6. What does a column on the periodic table represent?
7. What does a row on the periodic table represent?
8. What are the elements above the stepped line called?
9. How do the group 1 metals react with water?

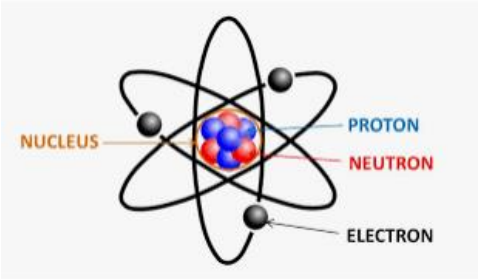
Useful links: https://www.youtube.com/watch?v=DZ6Ap8Zyb9w
https://www.youtube.com/watch?v=cpBb2bgFO6I
https://www.youtube.com/watch?v=0RRVV4Diomg

Chemistry Unit 1: Atomic Structure

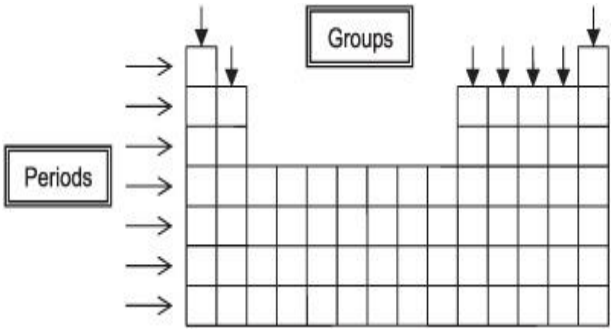
Key information to learn:

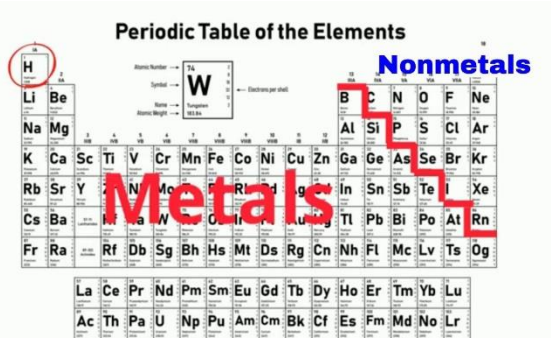


An atom is the smallest part of a chemical element.

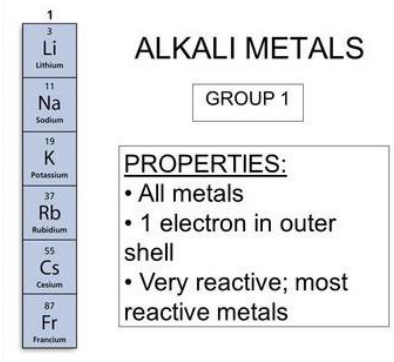


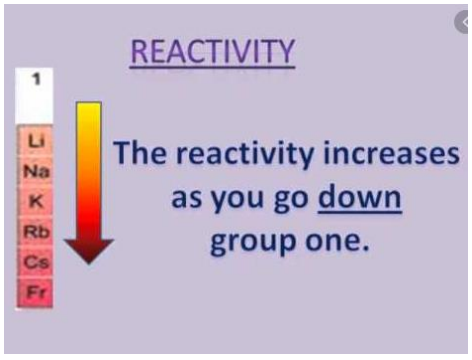
The periodic Table

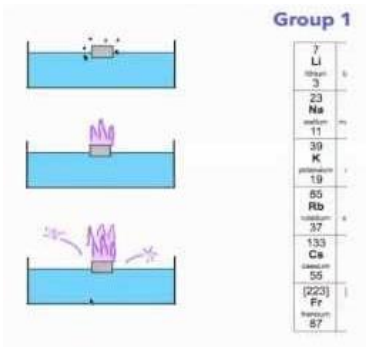




Group 1







Group 1 with water

Chemistry Unit 2: Bonding

Key points to learn:

1. How covalent bonds form.
2. Which elements covalent bonds form between.
3. How ionic bonds form.
4. Which elements ionic bonds form between.
5. How metallic bonds form.
6. Which elements metallic bonds form between.
7. The definition of an alloy.
8. The bonding found in alloys.
9. The definition of an allotrope.

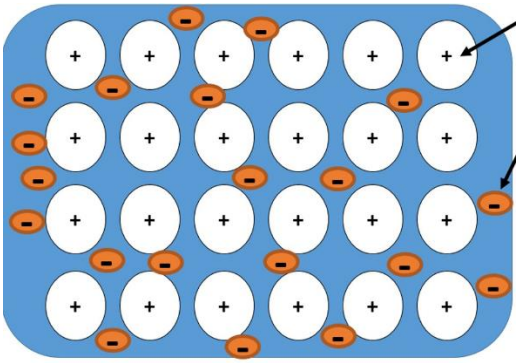
Practice Tasks/Questions

1. Do covalent bonds form by electrons being shared, transferred or delocalised?
2. Do ionic bonds form by electrons being shared, transferred or delocalised?
3. Do metallic bonds form by electrons being shared, transferred or delocalised?
4. Ionic bonds form between a metal and a non-metal. True or false?
5. Metallic bonds form between two non-metals. True or false?
6. What is an alloy?
7. What is an allotrope?
8. Write down the names of two allotropes of carbon.

Chemistry Unit 2: Bonding

Key information to learn:

- Covalent bonds are formed when non-metal atoms share electrons.
- Usually, each atom will donate one or two electrons to be shared.
- This gives 2 or 4 electrons being shared in total between the atoms.
- Ionic bonds are formed when a metal atom transfers an electron to a non-metal atom. This forms ions.
- Metallic bonds are formed when electrons become delocalised around metal ions.
- Delocalised means the electrons are free to move around (not stuck in one place).
- The diagram of metallic bonding is shown below.



- Alloys are a mixture of two or more elements where at least one of the elements is a metal.
- Alloys are often harder than pure metals.
- Allotropes are different forms of the same element.
- Graphite, graphene and diamond are all examples of allotropes made from carbon atoms.

Chemistry Unit 3: Chemical Analysis

Key points to learn:

1. The test for hydrogen.
2. The test for oxygen.
3. The test for carbon dioxide.
4. The test for chlorine.
5. What chromatography is.
6. The method for chromatography
7. What the common mistakes are for chromatography.
8. What colour flames are present in flame tests (J only)

Practice Tasks/Questions

1. How do you test for hydrogen? (test and positive result)
2. How do you test for oxygen? (test and positive result)
3. How do you test for carbon dioxide? (test and positive result)
4. How do you test for chlorine? (test and positive result)
5. What is chromatography used to separate?
6. Name 2 common practical mistakes made when setting up chromatography.
7. Write a method for completing the separation of food colourings by chromatography.
8. How do you complete a flame test.
9. Name 3 metal compounds and the colour they turn the flame

Useful links: https://www.youtube.com/watch?v=bcRGfSlMIMw
https://www.youtube.com/watch?v=TdJ57SQ6GAQ&t=1s
https://www.youtube.com/watch?v=1BCc_RrrSSw

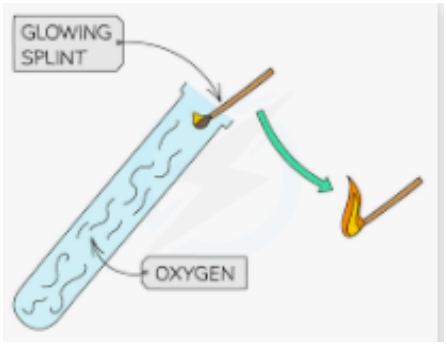
Chemistry Unit 3: Chemical Analysis

Key information to learn:

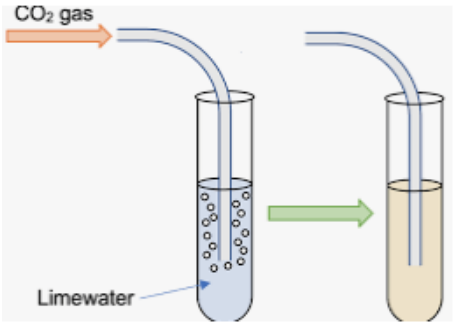
Test for hydrogen



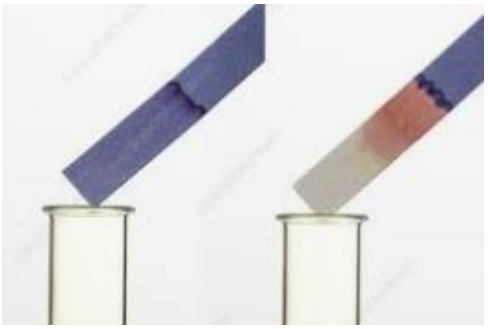
Test for oxygen



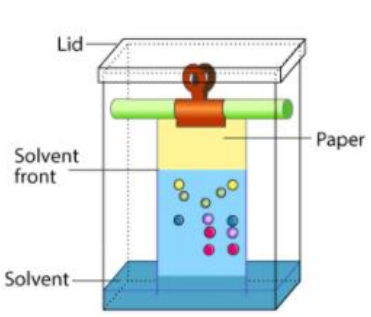
Test for carbon dioxide

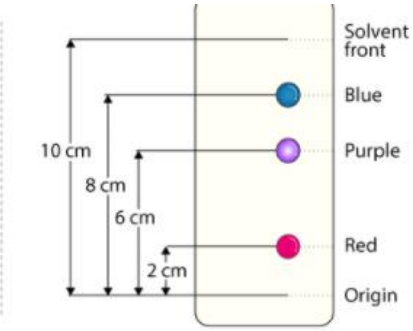


Test for chlorine



Chromatography



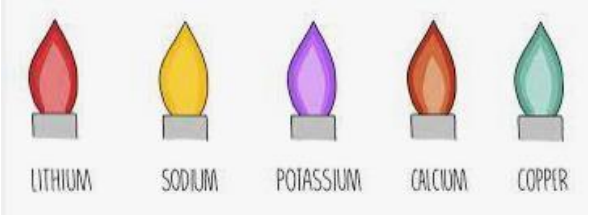


Chromatography Method

- Draw a line in pencil 1cm up from the bottom of the paper.
- A drop of mixture solution is spotted near one end of the paper.
- The end of the paper, nearest the spot, is then dipped into the solvent without submerging the spot itself.
- The results are then read when the solvent front nears the other end of the paper.

Common mistakes: using pen not pencil, not using the right solvent, submerging the sample spots.

Flame Test Results (J only)



Chemistry Unit 4: Rates

Key points to learn:

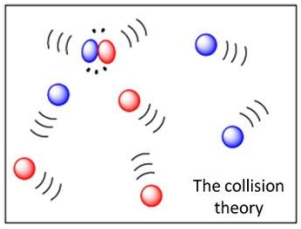
1. What is meant by rate of reaction, activation energy and collision theory.
2. Why chemical reactions take place.
3. Why increasing temperature increases rate of reaction.
4. Why increasing surface area increases rate of reaction.
5. Why increasing concentration increases rate of reaction.
6. What a catalyst is.
7. The effect catalysts have on rate of reaction.

Practice Tasks/Questions

1. What is collision theory?
2. What happens if two particles collide with less than the activation energy?
3. What 2 things need to happen for a reaction to take place?
4. What effect does decreasing temperature have on rate of reaction?
5. Why does increasing concentration increase rate of reaction?
6. What is meant by surface area?
7. Why is surface area important to rate of reaction?
8. What is a catalyst?
9. What effect do catalysts have on the activation energy of a chemical reaction?

Chemistry Unit 4: Rates

Key information to learn:

- The rate of chemical reaction is how quickly a reaction takes place.
- For a chemical reaction to take place the particles must collide (hit each other). They also must hit each other with a minimum amount of energy.
- 

- The minimum amount of energy particles must collide with for a reaction to happen is called the activation energy. If particles collide without the activation energy, then no chemical reaction happens.
- These are the two things that must happen for a reaction to take place: 1. The particles must collide; 2. They must collide with the activation energy.
- Increasing the temperature means the particles move around more quickly (more kinetic energy). As a result, they are more likely to collide and are more likely to have the activation energy. Both of these things mean increasing temperature increases the rate of a chemical reaction.
- Surface area is the area of an object exposed.
- When we increase surface area, we increase the area of the object that is exposed. This means there is more surface on which the particles can collide. This makes the particles more likely to collide and, as a result, rate of reaction increases when surface area increases.
- Concentration is the number of atoms, molecules or compounds in a given space.
- When concentration is increased there are more particles in the space. This means the particles are more likely to collide. Increases the frequency of collisions means the rate of reaction increases when concentration is increased.
- A catalyst is something that speeds up a chemical reaction without being used up.
- A catalyst increases the rate of a chemical reaction by lowering the activation energy. This means when the particles collide, they are more likely to have the activation energy and, therefore, rate of reaction increases.

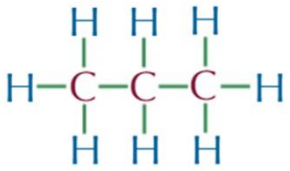
Chemistry Unit 5: Organic Chemistry

Key points to learn

1. How crude oil is formed.
2. How crude oil is separated using fractional distillation.
3. The elements in a hydrocarbon.
4. The names and structures of the first 4 alkanes.
5. How the properties of hydrocarbons change as the size increases.
6. What combustion is.
7. The products of a combustion reaction.
8. The difference between complete and incomplete combustion.

Practice Tasks/Questions

1. Describe how crude oil is formed.
2. In fractional distillation, do the small compounds gather at the top or bottom of the fractional distillation column?
3. What are the 2 elements in a hydrocarbon?
4. Which alkane is shown in this diagram?

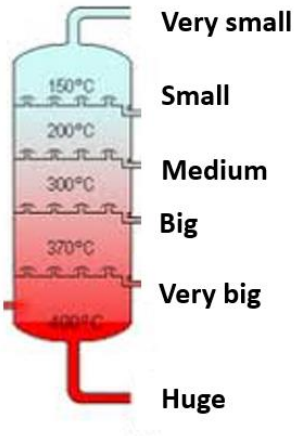


5. Draw the structure of ethane.
6. As the length of a hydrocarbon increases, what happens to:
   a. the viscosity?
   b. the flammability?
   c. the boiling point?
7. Which gas does a substance react with during combustion?
8. State the products of complete combustion.
9. State the products of incomplete combustion.

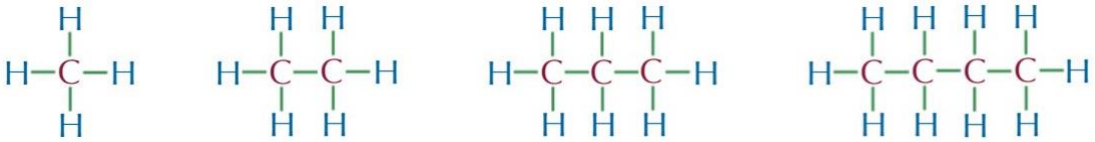
Chemistry Unit 5: Organic Chemistry

Key information to learn:

- Crude oil is formed from the remains of marine plants and animals. When they died, they fell to the bottom of the seabed and get covered by sediment. Over millions of years, heat and pressure turned the remains into crude oil.
- During fractional distillation, crude oil is heated up. This causes smaller fractions to rise higher than the larger fractions, which separates them out.



Methane *Ethane* *Propane* *Butane*



- As the length of the hydrocarbon or alkane increases…
  …the viscosity increases.
  …the flammability decreases.
  …the boiling point increases.
- Complete combustion:
  hydrocarbon + oxygen → carbon dioxide + water
- Incincomplete combustion:
  hydrocarbon + limited oxygen → carbon monoxide + water

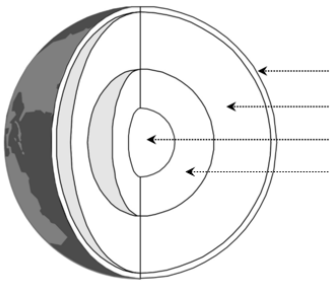
Keyword	**Definition**
Crude oil	Crude oil is a mixture of a very large number of compounds. Most of these compounds are hydrocarbons.
Hydrocarbon	A compound that only contains **hydrogen** and **carbon** atoms.
Viscosity	A measure of how thick a liquid is.
Flammability	A substance's ability to burn.
Boiling point	The temperature at which a liquid turns to a gas.
Combustion	The scientific word for burning.

Chemistry Unit 6: Chemistry of the Atmosphere

Key points to learn

1. The 4 main layers of the Earth.
2. The 3 types of rock.
3. How the rock cycle works.
4. How to extract metal from its metal ore.
5. The difference between Earth's early atmosphere and Earth's atmosphere today.
6. How the levels of carbon dioxide have changed over time.
7. The 3 greenhouse gases.
8. The effects of global warming.
9. How human activity affects global warming.

Practice Tasks/Questions

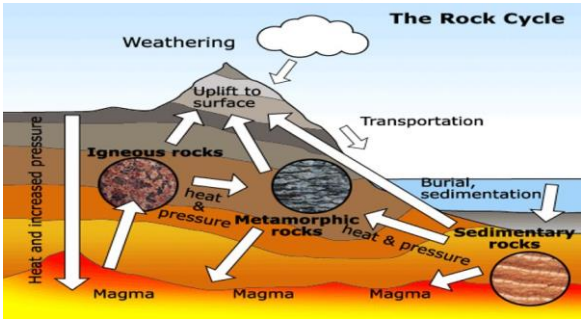


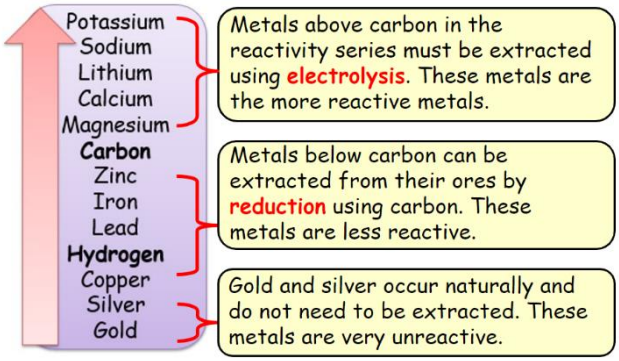
1. Label the 4 main layers of the Earth.
2. Describe the 4 main layers of the Earth.
3. Describe the 3 types of rock.
4. Explain how different types of rock can be transformed in the rock cycle.
5. Explain why calcium cannot be extracted from its ore by reduction.
6. Describe 2 ways that the levels of carbon dioxide in Earth's atmosphere were reduced.
7. What are the 3 greenhouse gases?
8. State 3 consequences of global warming.
9. State 2 ways that human activity increases the levels of greenhouse gases.

Chemistry Unit 6: Chemistry of the Atmosphere

Key information to learn:

- There are 4 main layers of the Earth:
  - crust: thin solid layer on the outside of the Earth
  - mantle: mostly solid rocks which can move slowly deep down
  - outer core: liquid layer, mostly made of iron and nickel
  - inner core: solid due to high pressure, mostly iron and nickel
- There are 3 types of rock:
  - sedimentary: formed from layers of sediment on sea and lake beds over millions of years, can contain fossils
  - igneous: formed from magma pushed to the surface of the crust, minerals form crystals
  - metamorphic: formed due to high pressure and heat acting on other rock types causing them to change structure





- Earth's atmosphere today is 80% nitrogen, 20% oxygen and <1% other gases.
- Earth's early atmosphere was mostly carbon dioxide. CO_2 levels reduced by dissolving in the oceans, being used in photosynthesis and being trapped in rocks.
- There are 3 greenhouse gases: carbon dioxide, methane and water vapour.
- Greenhouse gases absorb energy radiated from the Earth causing it to heat up. This is called global warming.
- Activities, such as burning fossil fuels and farming, release extra greenhouse gases.

Keyword	Definition
Ore	A rock that is a mixture of metals and other compounds.
Atmosphere	The gases surrounding a planet.

Physics Unit 1: Particles

Key points to learn:

1. The particle model diagram (how particles are arranged) for a solid, liquid and gas.
2. How particles behave in a solid, in a liquid and in a gas.
3. The names of each of the changes of state. (i.e. what do we call it when a solid changes to a liquid?)
4. The temperature that water freezes and boils at.
5. What causes pressure in gasses.
6. How pressure can be increased in a gas.
7. The definition of density.
8. How to calculate density.

Practice Tasks/Questions

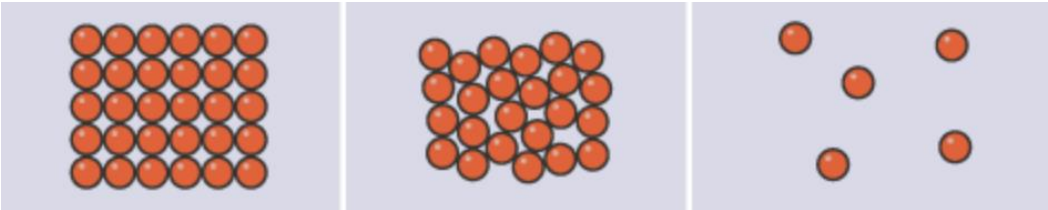
1. Draw the particle diagram for a solid, liquid and gas.
2. In which state(s) of matter do the particles vibrate?
3. In which state(s) of matter can the particles flow?
4. What do we call it when liquid turns to gas?
5. What do we call it when solid turns to liquid?
6. What causes pressure in gases?
7. Will increasing temperature increase or decrease the pressure of a gas?
8. Define density.
9. The mass of a block is 10 g. Its volume is 2 cm^3. Calculate the density of the block

Physics Unit 1: Particles

Key information to learn:

- The particle model for a solid, liquid and gas is shown below. You need to be able to draw these.

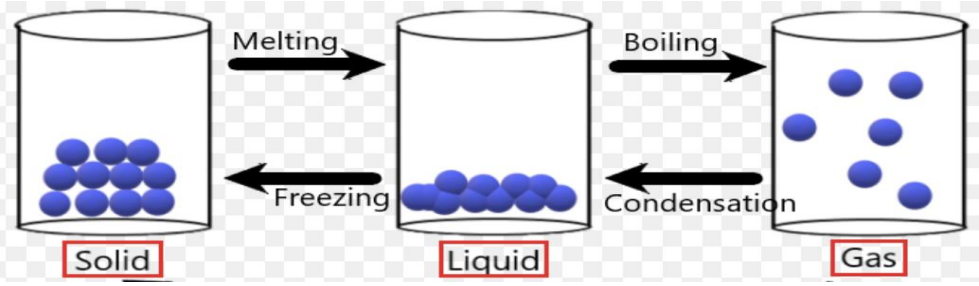
Solid | Liquid | Gas



- The table below summaries how particles behave in each of the states of matter.

State	Solid	Liquid	Gas
Closeness of particles	Very close	Close	Far apart
Arrangement of particles	Regular pattern	Randomly arranged	Randomly arranged
Movement of particles	Vibrate around a fixed position	Move around each other	Move quickly in all directions
Energy of particles	Low energy	Greater energy	Highest energy

- The diagram below shows the names of the changes of state. (i.e., it shows a liquid turning to a solid is called freezing.)



- Pressure in gases is caused when particles collide (hit) the walls of the container that they are in.
- Pressure can be increased by increasing temperature or the number of particles in the container.
- Density is a measure of how much 'stuff' or mass there is for a certain amount of space or volume.
- The equation for calculating density is given by $Density = \frac{mass}{volume}$

Physics Unit 2: Electricity

Key points to learn:

1. The charge of an electron.
2. How static electricity is formed.
3. How to draw electrical circuits.
4. What flows in a circuit.
5. The symbols of each of the components (parts) of a circuit.
6. The definition for a series circuit and a parallel circuit.
7. How current behaves in a series circuit.
8. How potential difference behaves in a series circuit.
9. How current behaves in a parallel circuit.
10. How potential difference behaves in a parallel circuit.

Practice Tasks/Questions

1. What is the charge on an electron?
2. How is static electricity formed?
3. Draw the circuit symbol for a bulb, cell and an ammeter.
4. What is the difference between a cell and a battery?
5. What flows in an electrical circuit?
6. Draw an electrical circuit to represent a series circuit with a battery, 2 bulbs and a switch.
7. State the difference between a series and a parallel circuit.
8. Is current the same or different at different parts of a series circuit?
9. Does current split or stay the same in a parallel circuit?
10. How does the potential difference compare along different branches of a parallel circuit? (Same/Different)

Physics Unit 2: Electricity

Key information to learn:

- Electrons are negatively charged.
- Materials can become statically charged due to friction. This moves the electrons from one material to another.
- A spark occurs when there is a flow of charge (current) through the air.
- Circuits must be drawn using a pencil and a ruler. They must have right angles corners, no gaps and no components on the corners.
- The circuit symbols are shown in the image below.

Switch = 

Cell = 

Battery = 

Lamp = 

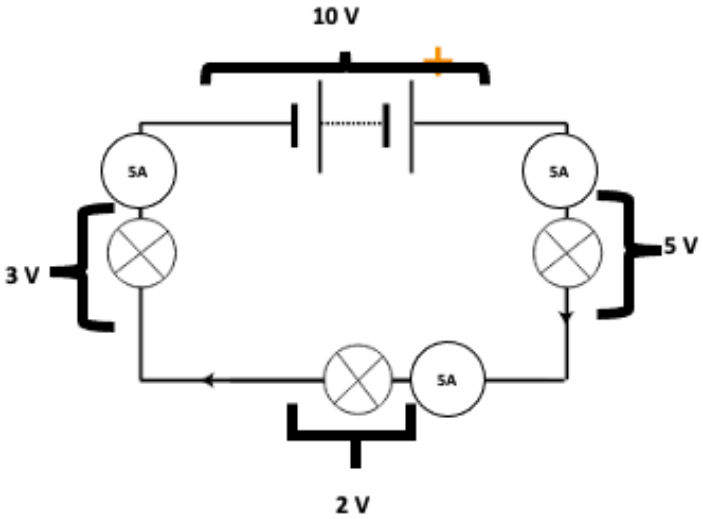
Voltmeter = 

Ammeter = 

Resistor = 

Motor = 

- A series circuit only has one loop. A parallel circuit has two or more loops.
- An ammeter is used to measure current.
- A voltmeter is used to measure potential difference.
- Current is the same at all points around a series circuit.
- Potential difference is shared between the different components of a series circuit. The circuit below shows how current and potential difference behave in series circuits.
- Current is shared down different branches of a parallel circuit.
- Potential difference is the same down difference branches of a parallel circuit.



Physics Unit 3: Energy

Key points to learn

Topic area	Key information
Energy stores	• Be able to identify the main energy stores and demonstrate how these can be transferred. • Explain the principle of conservation of energy.
Gravitational potential energy	• Understand that objects above the Earth have gravitational potential energy. • Use an equation to calculate gravitational potential energy.
Elastic potential energy	• Describe Elastic potential energy and demonstrate how elastic objects use this principle.
Efficiency	• Be able to understand that most energy transfers are not 100% energy efficient. Compare ways to calculate efficiency.
Energy resources	• Demonstrate an understanding of different energy resources. • Explain what is a non-renewable, renewable and sustainable source.

Practice Tasks/Questions

1. An object at rest on top of a shelf is likely to have what energy store?
2. Batteries are stores of?
3. Energy is usefully transferred from a personal audio player as?
4. Which type of energy transfer requires particles?
5. Which type of energy transfer can occur though a vacuum?
6. What type of energy is stored in the spring of a wind-up toy?
7. What type of diagram shows the efficiency of energy transfer?
8. What equation do we use to calculate gravitational potential energy?
9. What is efficiency?
10. What energy resources do we use to generate electricity?

Try this quiz:

https://docbrown.info/ks3physics/7ImcHP6.htm

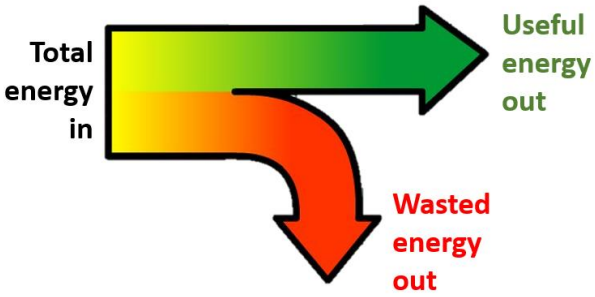
Physics Unit 3: Energy

Key information to learn:

Energy cannot be created or destroyed. Energy can only be **stored** or **transferred**.

Store	Description
Gravitational Potential	Energy stored in **high up** objects GPE = mass x gravitational field strength x height
Kinetic	Energy stored in **moving** objects Kinetic energy = 0.5 x mass x speed^2
Elastic Potential	Energy stored in **stretched or squashed** objects Elastic potential energy = 0.5 x spring constant x extension^2
Thermal	Energy stored in **hot** objects
Chemical	Energy stored in **chemical bonds** (like in food or batteries)
Nuclear	Energy stored **in the nucleus** of atoms
Magnetic	Energy stored in magnetic objects in a magnetic field
Electrostatic	Energy stored by **separated charges**

Sankey diagram



Efficiency

Not all of the energy supplied to a machine is transferred where we want it (usefully). Some energy goes where we don't want it to (wasted). The less energy is wasted, the more efficient the machine is.

$$efficiency = \frac{useful\ energy\ out}{total\ energy\ in} \times 100$$

Energy Resources

Humans use machines that need energy all the time (TVs, computers, fridges, cookers etc). We get the energy to run these machines from energy resources.

Non-renewable (will run out)	Renewable (constantly being replaced)
• Coal • Oil • Gas (Fossil fuels) • Nuclear	• Wind • Solar • Hydroelectric • Biomass

Check out these videos:

- Energy Stores
- Gravitational potential energy
- Elastic potential energy
- Efficiency
- Energy Resources

Physics Unit 4: Forces – Part 1

Key points to learn

1. What forces are and what three effects they can have on an object.
2. The equipment needed to measure forces.
3. The unit for forces.
4. How to draw force diagrams.
5. How to simplify a force diagram into a single resultant force.
6. The effect a resultant force can have on the acceleration of an object.
7. The difference between mass and weight.
8. The equation to calculate weight.
9. The relationship between force and extension of a spring.

Practice Tasks/Questions

1. State the three effects that forces can have on an object.
2. Name the piece of equipment used to measure forces.
3. Two tug of war teams pull on a rope. The blue teams pulls to the left with 800 N. The red team pull to the right with 800 N.

   800 N ←— → 800 N

   a. What is the resultant force on the rope?
   b. What will happen to the rope?
4. A box is still on the floor. A green man pushed to the left with 50 N. A purple man pushes to the right with 20 N.

   50 N ⟵ ❒ → 20 N

   a. What is the size and direction of the resultant force?
   b. What will happen to the box?
5. Calculate the weight of a bike with a mass of 50 kg. On Earth, the gravitational field strength, g = 10 N/Kg. **Show working!**
6. A spring obeys Hooke's Law. A force of 10 N is pulling the spring and it extends by 4 cm. The force is then doubled to 20 N. What will be the extension of the spring now?

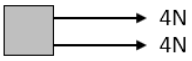
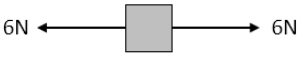
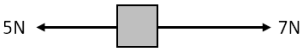
Physics Unit 4: Forces – Part 1

Key information to learn:



- Forces can affect an object in three ways:
  1. Change their speed 2. Change their direction 3. Change their shape
- Forces are measured using a Newton meter. This is a Newton meter →
- Forces are represented by arrows. The **length** of the arrow tells you about the **size** of the force. Longer arrow = bigger force. Forces also have a **direction**.
- The resultant force tells you what will happen to an object:
  - balanced forces = no resultant force → no acceleration
  - unbalanced forces = resultant force → object accelerates in direction of force
- The size of the force and the acceleration are linked by Newton's 2nd Law:

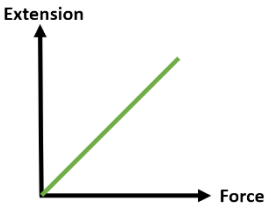
Force = mass x acceleration

Diagram	Resultant Force (size AND direction)	Effect on the motion of the box?
	Forces in same direction: add them. Resultant = 4 + 4 = 8N to the right	Resultant force right → box accelerates to the right
	Forces in opposite direction: take them away. Resultant = 6 - 6 = 0 N (no resultant)	No resultant force → no acceleration. The box stays still.
	Forces in opposite direction: take them away. Resultant = 7 - 5 = 2 N to the right (the biggest force wins)	Resultant force right → box accelerates to the right

- Mass and weight are NOT the same. Learn the definitions below. Your **mass stays the same** if you travel the solar system. Your **weight will change** depending on which planet you are on (as planets have different strengths of gravity).
- The equation to calculate weight:

weight = mass x gravitational field strength

- Springs obey Hooke's law. If you apply a force to a spring it extends (gets longer). If you double the force, then you will double the extension. We say that force and extension are directly proportional. A graph of force vs extension will be a straight line.



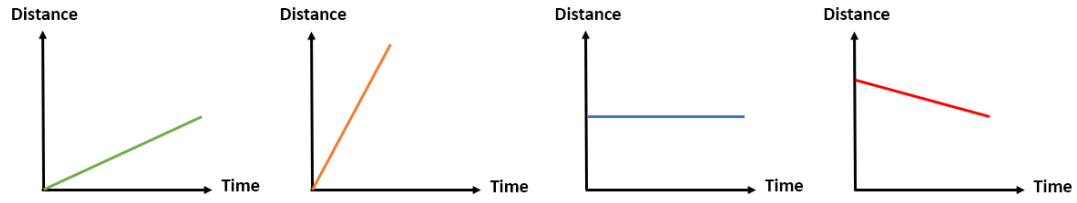
Keyword	Definition
accelerate	Changing the speed and/or the direction of travel of an object.
resultant	A single, imaginary force that would have the same effect as all the forces acting on an object combined.
mass	A measure of how much matter or "stuff" something is made of, measured in kilograms, kg.
weight	The force felt by a mass due to gravity.
gravitational field strength	A measure of how strong the pull of gravity is on a mass. On Earth, g is about 10 N/kg
extension	How much longer a spring is when stretched compared to its original length

Physics Unit 5: Forces – Part 2

Key points to learn

1. The relationship between force, pressure and area.
2. The equation used to calculate pressure.
3. How effects of pressure factor into the design of objects.
4. How pressure is caused in fluids.
5. The equation used to calculate speed.
6. How to interpret the shape of a distance vs time graph.
7. How to use the gradient of a distance vs time graph to calculate speed.
8. How to draw a distance vs time graph for a given journey.

Practice Tasks/Questions

1. What is the unit for pressure?
2. A crate has an area of 0.5 m^2. It applies a force of 650 N to the floor. Calculate the pressure of the crate on the floor.
3. Explain how the following objects have been designed to do their job well. Talk about pressure in your answers:
   a. A tank that doesn't sink in the mud.
   b. A pin that can be pushed easily into a noticeboard.
4. A helium filled balloon will float upwards if you let go of it. Explain why the balloon will expand as it gets higher, and eventually pop.
5. Calculate the speed of a bike that travels a distance of 910 m in a time of 65 seconds.
6. 

   Which graph shows:
   a. someone standing still
   b. someone travelling backwards?
   c. the person travelling forward the fastest?
7. Write a step-by-step guide for finding the speed of an object using the gradient of a distance vs time graph.
8. Sketch a distance vs time graph for someone who runs to a bus stop, waits for the bus, rides the bus to school, sits still in school then slowly walks home.

Physics Unit 5: Forces – Part 2

Key information to learn:

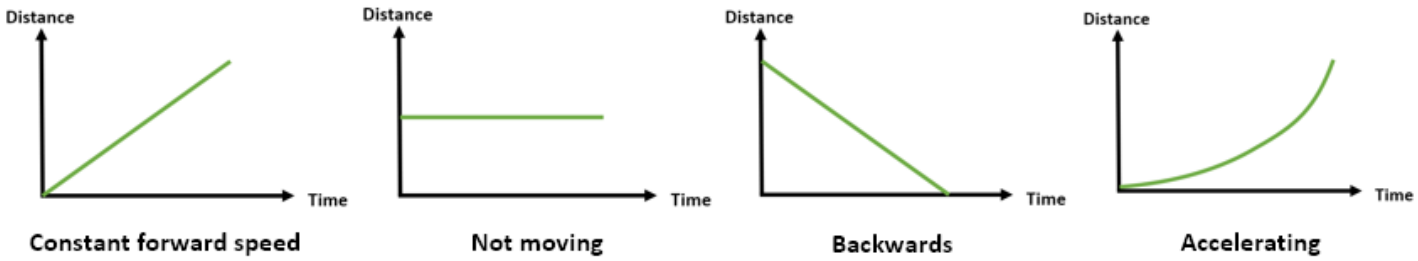
- Pressure is a measure of how spread out a force is. Pressure can be calculated using the equation:

pressure = force ÷ area

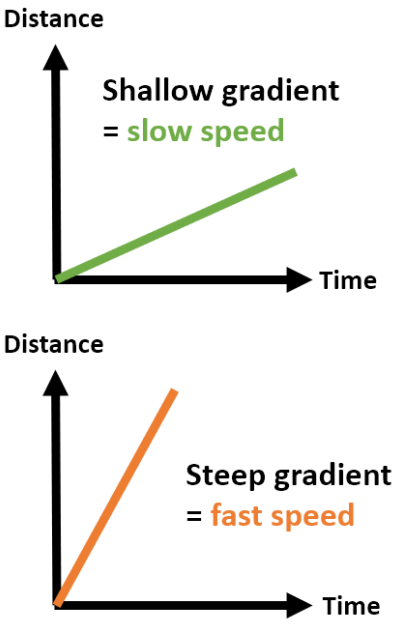
- Objects can be designed with pressure in mind:
  • low pressure – the force is spread over the large area of the pads in rucksack straps.
  • high pressure – the small area of a knife blade means it can cut easily.
- Pressure in fluids (like gases and liquids) acts in all directions and increases with depth.
- Speed can be calculated using the equation:

speed = distance ÷ time

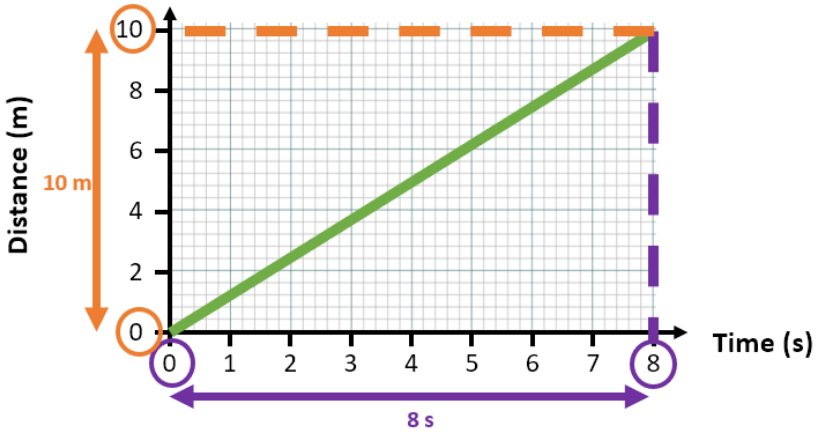
- The shape of a distance vs time graph can tell you about the journey:



- The gradient of a distance vs time graph can be used to find the speed of an object.



Finding speed from the gradient



speed = distance ÷ time
= 10 ÷ 8
= 1.25 m/s

Physics Unit 6: Magnetism

Key points to learn

1 Magnets and Magnetic Materials	• Describe how magnets behave • Explain how to produce an induced magnet • Explain why some alloys are magnetic
2 Magnetic Fields	• Draw the shape of magnetic fields. • Describe how to demonstrate the shape of a magnetic field. • Explain why the Earth has a magnetic field.
3 Electromagnets	• Describe the shape of the magnetic field around a current-carrying wire. • Describe how to make an electromagnet. • Describe how to increase the strength of an electromagnet.
5 Using Magnets	• Describe the uses of permanent magnets. • Describe the uses of temporary magnets. • Describe the advantages and disadvantages of temporary magnets.

Practice Tasks/Questions

1. Name the two magnetic poles?
2. Name the three magnetic elements and one magnetic alloy?
3. What do two poles of the same type do when they are brought close together?
4. What do two unlike poles do when they are brought close together?
5. The arrow on magnetic field lines shows them flowing in which direction?
6. Where are the field lines most concentrated around a bar magnet?
7. How can we increase the strength of an electromagnet?
8. What is one difference between electromagnets and bar magnets?
9. What can we use to find the shape of a magnetic field?
10. Which of these doesn't usually use an electromagnet?
    a. a scrap yard crane
    b. a fridge door magnet
    c. the magnet used in a recycling centre to sort cans

Physics Unit 6: Magnetism

Key information to learn:

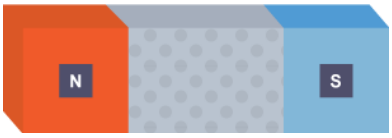
Bar magnets

Most materials are not magnetic, but some are. A magnetic material can be magnetised or will be attracted to a magnet. These metals are magnetic:

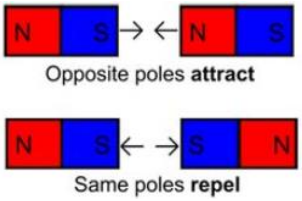
Elements: • Iron (Fe) • Cobalt (Co) • Nickel (Ni)

Alloy: • Steel is mostly iron, so steel is magnetic too.

A bar magnet is a permanent magnet. This means that its magnetism is there all the time and cannot be turned on or off. A bar magnet has two magnetic poles:



north pole (or north-seeking pole)
south pole (or south-seeking pole)



Attract and repel

If you bring two bar magnets together, there are two things that can happen, attraction and repulsion:
We say that opposite poles attract, and like poles repel.

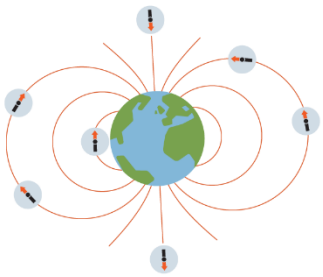
Finding magnetic fields

You can use a plotting compass or iron filings to detect a magnetic field:

- put a piece of paper over a magnet (this stops the iron filings sticking to the magnet)
- sprinkle iron filings onto the paper
- gently tap the paper to spread the filings out
- observe and record the results



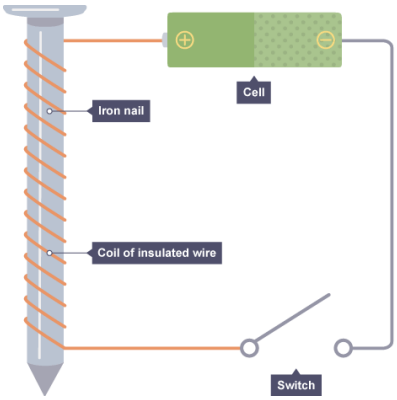
The Earth's magnetism



The Earth behaves as if it contains a giant magnet. It produces a magnetic field in which the field lines are most concentrated at the poles. This magnetic field can be detected using magnetic materials or magnets.

Electromagnets

When an electric current flows in a wire, it creates a magnetic field around the wire. This effect can be used to make an electromagnet. A simple electromagnet comprises a length of wire turned into a coil and connected to a battery or power supply.



Revision Video https://youtu.be/yXCeuSiTOug